no duplicate tuples. Using the **unique** construct, we can write the query "Find all customers who have at most one account at the Perryridge branch" as follows:

```
select T.customer-name
from depositor as T
where unique (select R.customer-name
              from account, depositor as R
              where T.customer-name = R.customer-name and
                    R.account-number = account.account-number and
                    account.branch-name = 'Perryridge')
```

We can test for the existence of duplicate tuples in a subquery by using the **not unique** construct. To illustrate this construct, consider the query "Find all customers who have at least two accounts at the Perryridge branch," which we write as

```
select distinct T.customer-name
from depositor T
where not unique (select R.customer-name
                  from account, depositor as R
                  where T.customer-name = R.customer-name and
                        R.account-number = account.account-number and
                        account.branch-name = 'Perryridge')
```

Formally, the **unique** test on a relation is defined to fail if and only if the relation contains two tuples t_1 and t_2 such that $t_1 = t_2$. Since the test $t_1 = t_2$ fails if any of the fields of t_1 or t_2 are null, it is possible for **unique** to be true even if there are multiple copies of a tuple, as long as at least one of the attributes of the tuple is null.

4.7 Views

We define a view in SQL by using the **create view** command. To define a view, we must give the view a name and must state the query that computes the view. The form of the **create view** command is

create view v **as** <query expression>

where <query expression> is any legal query expression. The view name is represented by v. Observe that the notation that we used for view definition in the relational algebra (see Chapter 3) is based on that of SQL.

As an example, consider the view consisting of branch names and the names of customers who have either an account or a loan at that branch. Assume that we want this view to be called *all-customer*. We define this view as follows: